

SAPTHA SANKA GAVESH COSTA

Software Engineering Undergraduate

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PROFESSIONAL SUMMARY

Software Engineering undergraduate seeking an Internship or Associate Software Engineer role. Builds multi-tenant SaaS, AI and edge computer vision systems using React, Node.js, PostgreSQL and Python, with IBM, Meta and Google professional certificates. BSc (Hons) Computing (Software Engineering) in progress at AIBS, awarded by Edge Hill University, UK.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C++
Front-End: React, Angular, HTML5, CSS3, Tailwind CSS, Responsive Design
Back-End: Node.js, Express, Next.js, FastAPI, Spring Boot, REST APIs, WebSockets
Databases: PostgreSQL, Drizzle ORM, Supabase, MySQL, MongoDB, Redis, ChromaDB (vector database)
AI & Machine Learning: PyTorch, YOLOv8 (Computer Vision), TensorFlow, NLP, LangGraph, TinyML, Prompt Engineering
DevOps & Cloud: Docker, Kubernetes, Terraform, AWS, GitHub Actions, CI/CD, OpenTelemetry, Prometheus, Grafana, n8n
Testing & QA Automation: Selenium, Katalon Studio, Cypress, Jest, Unit & Integration Testing, Postman
Version Control: Git, GitHub
AI-Assisted Development: Claude Code, Cursor, Lovable, Bolt, Replit — rapid prototyping

PROJECT EXPERIENCE

ReconLK — Enterprise Financial OS & Multi-Ledger Reconciliation

Mar 2026 – Present

React 19, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM GitHub: github.com/sanka-dev425/projectrecon

- Designed a multi-tenant financial operating system featuring double-entry bookkeeping guarantees, ACID transaction invariants, and PostgreSQL Row-Level Security (RLS) for tenant isolation.
- Built an automated high-throughput transaction matching engine capable of auditing multi-ledger discrepancies and executing background reconciliation jobs without blocking core operations.
- Developed a responsive dashboard using React 19, TypeScript, and TailwindCSS providing real-time financial telemetry, automated POS invoicing, and audit-ready reporting.

ArgusTraffic AI — Autonomous Edge Vision & Spatial Intelligence Engine

Python, PyTorch, YOLOv8, FastAPI, Redis, Docker GitHub: github.com/sanka-dev425/argustraffic-ai

- Architected an edge computer vision platform ingesting multi-stream RTSP/UVC feeds for spatial vehicle tracking, optical velocity radar, and collision risk anticipation at sub-50ms inference latency.
- Engineered an asynchronous FastAPI micro-kernel decoupling high-throughput neural inference from WebSocket telemetry streams, accompanied by a SHA-256 Merkle chain aligned with ISO/IEC 27037 for forensic non-repudiation.
- Built and validated a 110/110 passing automated test suite with Docker containerization for cross-platform edge deployments.

Incident RCA Platform — Distributed Systems AI Incident Copilot

Python, FastAPI, LangGraph, OpenTelemetry, ChromaDB, Terraform GitHub: github.com/sanka-dev425/incident-rca-platform

- Developed an autonomous SRE copilot orchestrating multi-agent graph workflows (LangGraph) to diagnose distributed microservice anomalies and accelerate incident triage.
- Correlated OpenTelemetry distributed traces, Prometheus metrics, and Grafana Loki logs with a vector database (ChromaDB) of historical postmortems to generate root-cause hypotheses with confidence scoring.
- Implemented a fault-injection chaos harness and provisioned modular AWS cloud infrastructure via Terraform (IaC) with automated CI/CD quality gates.

AI Code Reviewer — Automated GitHub PR Review Action
TypeScript, Node.js, GitHub Actions API, Anthropic Claude, Jest GitHub: github.com/sanka-dev425/ai-code-reviewer

- Engineered a GitHub Action utilizing Anthropic Claude to analyze pull request diffs and publish automated, inline code review comments directly on modified lines.
- Integrated security-first review heuristics enforcing OWASP guidelines, detecting anti-patterns, evaluating test coverage, and establishing automated merge gates.
- Implemented token-cost optimization, idempotency safeguards (deleting stale review comments on re-run), and comprehensive Jest unit testing.

AI-Driven Multilingual Insurance Recommendation System *Prototype*
Python, NLP, Machine Learning, Explainable AI (XAI)

- Designed, coded, and trained a multilingual AI prototype delivering personalised life and health insurance recommendations in Sinhala, Tamil, and English, addressing Sri Lanka's financial accessibility gap.
- Implemented Explainable AI (XAI) techniques to provide clear, human-readable reasoning behind every recommendation, building user trust in AI-driven financial advisory tools.
- Built NLP pipelines to interpret multilingual user input and map it to optimal insurance products through trained machine learning models.
- Targeted a documented market failure: low insurance penetration in Sri Lanka driven by language barriers and lack of accessible guidance.

EDUCATION

- BSc (Hons) in Computing, Software Engineering — In Progress, Expected 2026/2027
  Awarded by Edge Hill University, UK. Delivered at Asian Institute of Business & Science (AIBS), Colombo 08, Sri Lanka
- Pearson BTEC Level 5 Higher National Diploma (HND) in Computing, Software Engineering — Completed 2026
  Asian Institute of Business & Science, Colombo 08, Sri Lanka

PROFESSIONAL CERTIFICATIONS

Cloud & DevOps

- Certified Kubernetes Application Developer (CKAD) Specialization — Pearson
- IBM DevOps and Software Engineering Professional Certificate — IBM

Software Engineering & Full-Stack Development

- IBM Full Stack Software Developer Professional Certificate — IBM
- Meta Full Stack Developer Professional Certificate — Meta
- Front-End Web Development — University of Moratuwa
- Server-Side Web Programming — University of Moratuwa

AI & Machine Learning

- IBM AI Engineering Professional Certificate — IBM
- Deep Learning Specialization — DeepLearning.AI
- Google AI Professional Certificate — Google

Quality Assurance & Test Automation

- Software QA & Test Automation Engineering Specialization — Coursera

Full portfolio: credly.com/users/saptha-sanka

VOLUNTEERING

- Scout Master — Sri Lanka Air Force Air Scout Group

PERSONAL DETAILS

Date of Birth: 9 February 2005
Nationality: Sri Lankan
Civil Status: Single
Languages: Sinhala and English